SEVEN EQUIVALENT CHARACTERIZATIONS OF POSSESSING AN \'ETALE-SMOOTH LOCAL ON SOURCE-AND-TARGET PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two definitions and one same-chapter lemma (carded locally), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- \'Etale-smooth local on source-and-target: definition. Stacks Project, Descent on Spaces, Definition definition-etale-smooth-local-source-target. Supplies the (a)<=>(b) equivalence.
- Permanence properties of \'etale-smooth local properties. Stacks Project, Descent on Spaces, Lemma lemma-etale-smooth-local-source-target-implies (carded locally as S11-spaces-desc-etale-smooth-implies). Supplies (a)<=>(e) and (a)<=>(f) via its parts (2) and (1).
- Smooth local on source-and-target: definition. Stacks Project, Descent on Spaces, Definition definition-local-source-target. Supplies smooth locality on the target used when coproducting the charts.

Source reference. Original definition 0CFZ: sources/stacks-chow/tags.txt, source lines 15629–15629. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CG0: sources/stacks-chow/tags.txt, source lines 15630–15630. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S which is etale-smooth local on source-and-target. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(a) *f has property $\mathcal{P}$,*
(b) *for every $x \in |X|$ there exists a smooth morphism $b : V \to Y$, an étale morphism $a : U \to V \times_Y X$, and a point $u \in |U|$ mapping to x such that $U \to V$ has $\mathcal{P}$,*
(c) *for some commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with b smooth, $U \to V \times_Y X$ étale, and a surjective the morphism h has $\mathcal{P}$,

Date: stacks-project:spaces-descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(d) *for any commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with b smooth and $U \to X \times_Y V$ étale, the morphism h has $\mathcal{P}$,

(e) *there exists a smooth covering $\{Y_i \to Y\}_{i \in I}$ such that each base change $Y_i \times_Y X \to Y_i$ has $\mathcal{P}$,*

(f) *there exists an étale covering $\{X_i \to X\}_{i \in I}$ such that each composition $X_i \to Y$ has $\mathcal{P}$,*

(g) *there exists a smooth covering $\{Y_i \to Y\}_{i \in I}$ and for each $i \in I$ an étale covering $\{X_{ij} \to Y_i \times_Y X\}_{j \in J_i}$ such that each morphism $X_{ij} \to Y_i$ has $\mathcal{P}$.*

Proof. The equivalence of (a) and (b) is part of Definition 0CFZ. The equivalence of (a) and (e) is Lemma 0CG0 part (2). The equivalence of (a) and (f) is Lemma 0CG0 part (1). As (a) is now equivalent to (e) and (f) it follows that (a) equivalent to (g).

It is clear that (c) implies (b). If (b) holds, then for any $x \in |X|$ we can choose a smooth morphism a smooth morphism $b_x : V_x \to Y$, an étale morphism $U_x \to V_x \times_Y X$, and $u_x \in |U_x|$ mapping to x such that $U_x \to V_x$ has $\mathcal{P}$. Then $h = \coprod h_x : \coprod U_x \to \coprod V_x$ with $a = \coprod a_x$ and $b = \coprod b_x$ is a diagram as in (c). (Note that h has property $\mathcal{P}$ as $\{V_x \to \coprod V_x\}$ is a smooth covering and $\mathcal{P}$ is smooth local on the target.) Thus (b) is equivalent to (c).

Now we know that (a), (b), (c), (e), (f), and (g) are equivalent. Suppose (a) holds. Let U, V, a, b, h be as in (d). Then $X \times_Y V \to V$ has $\mathcal{P}$ as $\mathcal{P}$ is stable under smooth base change, whence $U \to V$ has $\mathcal{P}$ as $\mathcal{P}$ is stable under precomposing with étale morphisms. Conversely, if (d) holds, then setting $U = X$ and $V = Y$ we see that f has $\mathcal{P}$. □